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Dielectric & Polarization

by Tony Shing

Overview:

- Electric dipole: Origin and basic model
- How to describe dipole arrangement: Polarization field $\vec{\boldsymbol{P}}$, bound charges ρ_b, σ_b
- Material under external $\vec{\boldsymbol{E}}$ field:
  - A special case of material model: Linear dielectric
  - Describe external field: Free charges ρ_f, σ_f, electric displacement field $\vec{\boldsymbol{D}}$

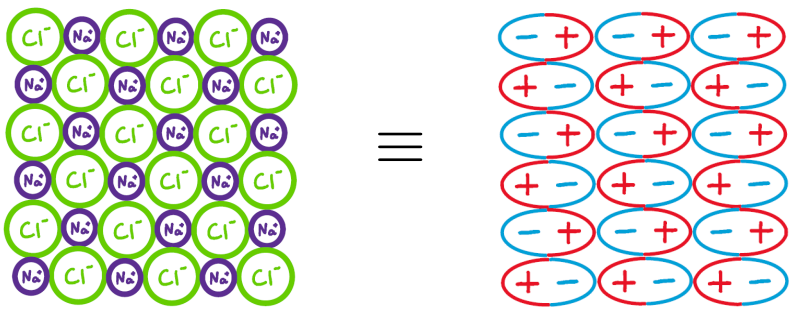
1 Electric Dipole

1.1 Insulators as Mixture of Dipoles

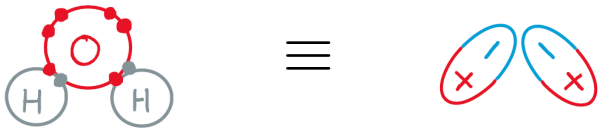
Every atom is intrinsically an electric dipole since its electrons' positions can fluctuate. In Chemistry class, this is called **instantaneous dipole**, because the direction and magnitude of the dipole vary over time.

Besides, chemical bondings between different types of atom can also form **permanent dipole**. The difference in atoms' electronegativity (i.e. strength of attracting outer shell electrons) causes uneven charge distribution around the atoms. These types of bondings exist only in insulator,i.e. **dielectric**. For example,

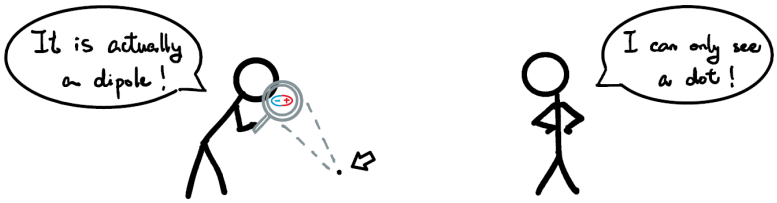
- <u>Ionic bond</u> - One atom completely takes an electron from another, forming a pair of positive and negative ions.



– (Polar) Covalent bond - Bonding electrons are shared by two atoms, but the atom with higher electronegativity pulls the electrons closer to itself, forming a polar covalent bond.



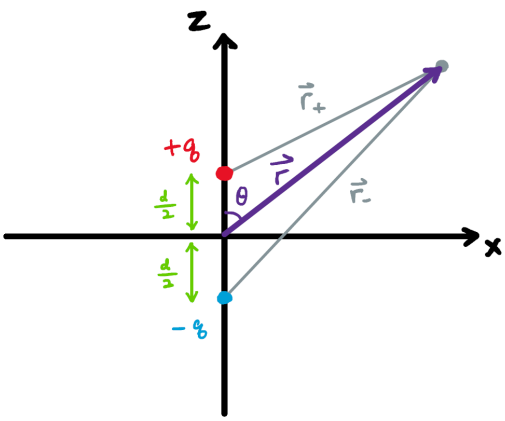
Since electric dipoles emit E-field, theoretically all dielectrics are polar. However the dipoles in real life are so small ($\sim 10^{-10}$ m) and are usually packed in random orientations. The net field can usually be treated as zero if we view the dipoles from a very large distance.



So polarization of big objects are normally unobservable, unless we apply an external E-field to align the dipoles in some direction.

1.2 Fields of Electric Dipole

Since electric dipole is the basic unit of dielectric, we can first analyze the field emitted by a single dipole, then sum the contributions from all dipoles according to the material structure. The classical model of electric dipole is as follows:



– Both ends of the dipole have equal but opposite *point* charge $+q$ and $-q$.

– Separation between the charges is labelled by a vector $\vec{\boldsymbol{d}}$, which points from the negative charge to the positive charge.

– We are only interested in the potential/field far away from the dipole, i.e. $|\vec{\boldsymbol{r}}| \gg |\vec{\boldsymbol{d}}|$. The position vector $\vec{\boldsymbol{r}}$ is referenced from the center of the dipole.

1.2.1 Potential by Electric Dipole

Deriving electric potential from a dipole is straightforward by Coulomb's law, while applying Taylor approximation in the middle step.

1. The total potential contributions from the two charges towards position $\vec{\boldsymbol{r}}$ is

$$
\begin{aligned}
V(\vec{\boldsymbol{r}}) &= \frac{1}{4\pi\epsilon_0}\left(\frac{q}{|\vec{\boldsymbol{r}}_+|} + \frac{-q}{|\vec{\boldsymbol{r}}_-|}\right) \\
&= \frac{q}{4\pi\epsilon_0}\left(\underbrace{\frac{1}{\sqrt{|\vec{\boldsymbol{r}}|^2 + \left|\frac{\vec{\boldsymbol{d}}}{2}\right|^2 - 2|\vec{\boldsymbol{r}}|\left|\frac{\vec{\boldsymbol{d}}}{2}\right|\cos\theta}}} - \underbrace{\frac{1}{\sqrt{|\vec{\boldsymbol{r}}|^2 + \left|\frac{\vec{\boldsymbol{d}}}{2}\right|^2 + 2|\vec{\boldsymbol{r}}|\left|\frac{\vec{\boldsymbol{d}}}{2}\right|\cos\theta}}}\right) \\
&\qquad\qquad \text{These are just cosine law} \\
&= \frac{q}{4\pi\epsilon_0|\vec{\boldsymbol{r}}|}\left(\frac{1}{\sqrt{1 + \frac{1}{4}\frac{|\vec{\boldsymbol{d}}|^2}{|\vec{\boldsymbol{r}}|^2} - \frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta}} - \frac{1}{\sqrt{1 + \frac{1}{4}\frac{|\vec{\boldsymbol{d}}|^2}{|\vec{\boldsymbol{r}}|^2} + \frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta}}\right)
\end{aligned}
$$

2. Taylor expansion "$(1+x)^n \approx 1 + nx$ for $x \ll 1$" is always used at this step. We only care about the potential at a distance much larger than the dipole's size, so take $\frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|} \ll 1$,

$$\Rightarrow \frac{1}{4}\frac{|\vec{\boldsymbol{d}}|^2}{|\vec{\boldsymbol{r}}|^2} \approx 0$$

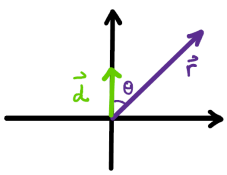
$$\Rightarrow \frac{1}{\sqrt{1 - \frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta}} = \left(1 - \frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta\right)^{-\frac{1}{2}} \approx 1 + \frac{1}{2}\cdot\frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta$$

So the potential becomes

$$
\begin{aligned}
V(\vec{\boldsymbol{r}}) &= \frac{q}{4\pi\epsilon_0|\vec{\boldsymbol{r}}|}\left[\left(1 + \frac{1}{2}\frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta\right) - \left(1 - \frac{1}{2}\frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta\right)\right] \\
&= \frac{q}{4\pi\epsilon_0|\vec{\boldsymbol{r}}|}\frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|}\cos\theta
\end{aligned}
$$

3. From geometry, θ is the angle between $\vec{\boldsymbol{d}}$ and $\vec{\boldsymbol{r}}$. We can replace $\cos\theta$ by $\cos\theta = \frac{\vec{\boldsymbol{d}}\cdot\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{d}}||\vec{\boldsymbol{r}}|}$,

$$
\begin{aligned}
V(\vec{\boldsymbol{r}}) &= \frac{q}{4\pi\epsilon_0}\frac{|\vec{\boldsymbol{d}}|}{|\vec{\boldsymbol{r}}|^2}\frac{\vec{\boldsymbol{d}}\cdot\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{d}}||\vec{\boldsymbol{r}}|} \\
&= \frac{1}{4\pi\epsilon_0}\frac{q\vec{\boldsymbol{d}}}{|\vec{\boldsymbol{r}}|^2}\underbrace{\left(\frac{\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|}\right)}_{\text{Unit vector of } \vec{\boldsymbol{r}}}
\end{aligned}
$$



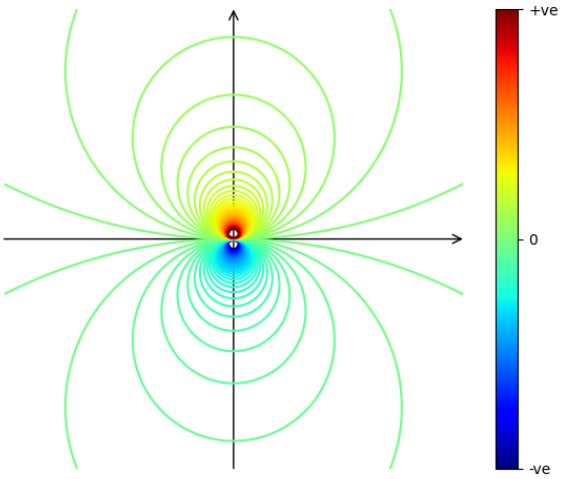
4. To emphasize that we now treat an electric dipoles as "one unit of source", We define the **electric dipole moment** $\vec{\boldsymbol{p}}$,

$$\boxed{\vec{\boldsymbol{p}} \overset{\text{def}}{=} q\vec{\boldsymbol{d}}}$$

Substitute to become the standard formula of electric potential from an electric dipole:

$$\boxed{V(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\frac{\vec{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|^2}}$$

(Lines in the plot = Equi-potentials)



For reference, we can see its similarities with Coulomb's law for point charge:

Magnitude by q

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{|\vec{\boldsymbol{r}}|}$$

Nothing directional

Proportionality constant

Decay by $\frac{1}{r}$

If the source is point charge

Magnitude by $|\vec{\boldsymbol{p}}|$

$$V = \frac{1}{4\pi\epsilon_0}\frac{\vec{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|^2}$$

Directional by angle between $\vec{\boldsymbol{p}}$ and $\hat{\boldsymbol{r}}$ (dot product)

Proportionality constant

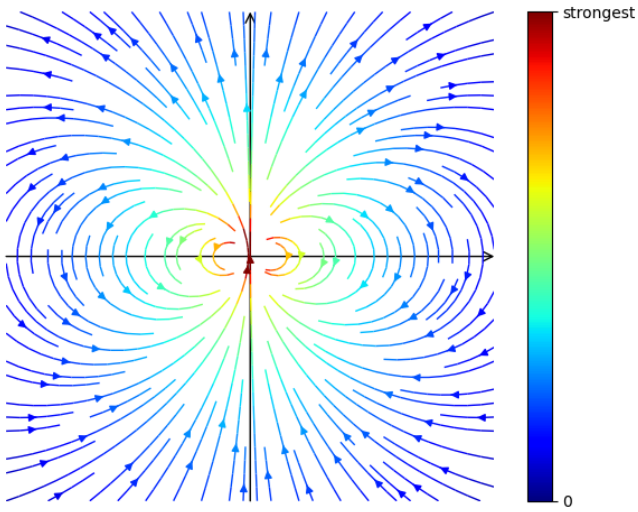
Decay by $\frac{1}{r^2}$

If the source is dipole

1.2.2 E-field by Electric Dipole

The standard derivation of E-field from dipole is through the relation $\vec{\boldsymbol{E}} = -\vec{\boldsymbol{\nabla}}V$. It is nothing more than some boring calculus (see next page). Here I just quote the final result:

$$\boxed{\vec{\boldsymbol{E}}(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\frac{3(\vec{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \vec{\boldsymbol{p}}}{|\vec{\boldsymbol{r}}|^3}}$$



And this formula is rarely used because vector calculation is annoying.

The boring derivation of E-field from dipole:

1. Derivation is possible with high school calculus if we turn all vectors into x-y-z form: $\vec{\boldsymbol{r}} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}$, $|\vec{\boldsymbol{r}}| = \sqrt{x^2+y^2+z^2}$ and $\vec{\boldsymbol{p}} = p_x\hat{\boldsymbol{x}} + p_y\hat{\boldsymbol{y}} + p_z\hat{\boldsymbol{z}}$:

$$\frac{\vec{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|^2} = \frac{\vec{\boldsymbol{p}}}{|\vec{\boldsymbol{r}}|^2}\cdot\frac{\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|} = \frac{1}{|\vec{\boldsymbol{r}}|^3}(\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}}) = \frac{1}{(x^2+y^2+z^2)^{\frac{3}{2}}}(p_x x + p_y y + p_z z)$$

2. First the x term in gradient, $\hat{\boldsymbol{x}}\dfrac{\partial}{\partial x}(\cdots)$, can be calculated by:

$$\hat{\boldsymbol{x}}\frac{\partial}{\partial x}\left[\frac{1}{(x^2+y^2+z^2)^{\frac{3}{2}}}(p_x x + p_y y + p_z z)\right]$$

This is just quotient rule

$$= \frac{\hat{\boldsymbol{x}}}{(x^2+y^2+z^2)^3}\left[p_x(x^2+y^2+z^2)^{\frac{3}{2}} - (p_x x + p_y y + p_z z)\cdot 3x(x^2+y^2+z^2)^{\frac{1}{2}}\right]$$

$$= \frac{1}{(x^2+y^2+z^2)^{\frac{5}{2}}}\left[p_x\hat{\boldsymbol{x}}\underline{(x^2+y^2+z^2)} - \underline{(p_x x + p_y y + p_z z)}\cdot 3x\hat{\boldsymbol{x}}\right]$$

$$= \frac{1}{|\vec{\boldsymbol{r}}|^5}\left[p_x\hat{\boldsymbol{x}}\underline{|\vec{\boldsymbol{r}}|^2} - 3\underline{(\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}})}x\hat{\boldsymbol{x}}\right]$$

3. Similar for y and z terms:

$$\hat{\boldsymbol{y}}\frac{\partial}{\partial y}(\cdots) = \frac{1}{|\vec{\boldsymbol{r}}|^5}\left[p_y\hat{\boldsymbol{y}}|\vec{\boldsymbol{r}}|^2 - 3(\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}})y\hat{\boldsymbol{y}}\right]$$

$$\hat{\boldsymbol{z}}\frac{\partial}{\partial z}(\cdots) = \frac{1}{|\vec{\boldsymbol{r}}|^5}\left[p_z\hat{\boldsymbol{z}}|\vec{\boldsymbol{r}}|^2 - 3(\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}})z\hat{\boldsymbol{z}}\right]$$

4. Summing all 3 components gives:

$$-\vec{\boldsymbol{\nabla}}V(\vec{\boldsymbol{r}}) = -\left(\hat{\boldsymbol{x}}\frac{\partial V}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial V}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial V}{\partial z}\right)$$

$$= -\frac{1}{4\pi\epsilon_0}\frac{1}{|\vec{\boldsymbol{r}}|^5}\left[\underline{(p_x\hat{\boldsymbol{x}} + p_y\hat{\boldsymbol{y}} + p_z\hat{\boldsymbol{z}})}|\vec{\boldsymbol{r}}|^2 - 3(\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}})\underline{(x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}})}\right]$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{|\vec{\boldsymbol{r}}|^5}\left[3(\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}})\underline{\vec{\boldsymbol{r}}} - \underline{\vec{\boldsymbol{p}}}|\vec{\boldsymbol{r}}|^2\right]$$

Take out magnitude to become unit vector

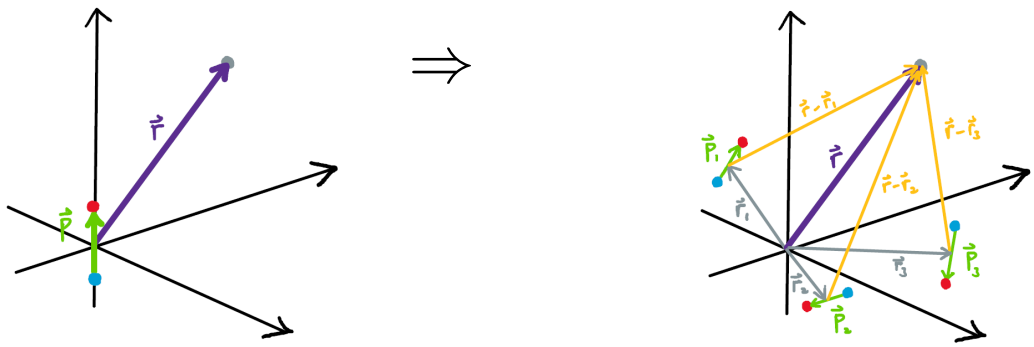
$$= \frac{1}{4\pi\epsilon_0}\frac{1}{|\vec{\boldsymbol{r}}|^5}\left[3(\vec{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}}\,|\vec{\boldsymbol{r}}|^2 - \vec{\boldsymbol{p}}|\vec{\boldsymbol{r}}|^2\right]$$

$$\boxed{\vec{\boldsymbol{E}}(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\frac{3(\vec{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \vec{\boldsymbol{p}}}{|\vec{\boldsymbol{r}}|^3}}$$

2 Describing Dipole Arrangement

In general, all dielectric materials are made of many tiny electric dipoles. To analyze the material's electric properties, we can begin with the property of a single dipole, then sum the contributions of all dipoles according to the dipole arrangement in the material.

$$\underbrace{V(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\frac{\vec{\boldsymbol{p}}}{|\vec{\boldsymbol{r}}|^2}\cdot\left(\frac{\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|}\right)}_{\text{By a single dipole from origin}} \qquad \underbrace{V(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\sum_{\text{All dipoles } i}\frac{\vec{\boldsymbol{p}}_i}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i|^2}\cdot\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i|}\right)}_{\text{By many dipoles at different positions } \vec{\boldsymbol{r}}_i}$$



Now we introduce two quantities that can be used to describe electric dipole arrangements.

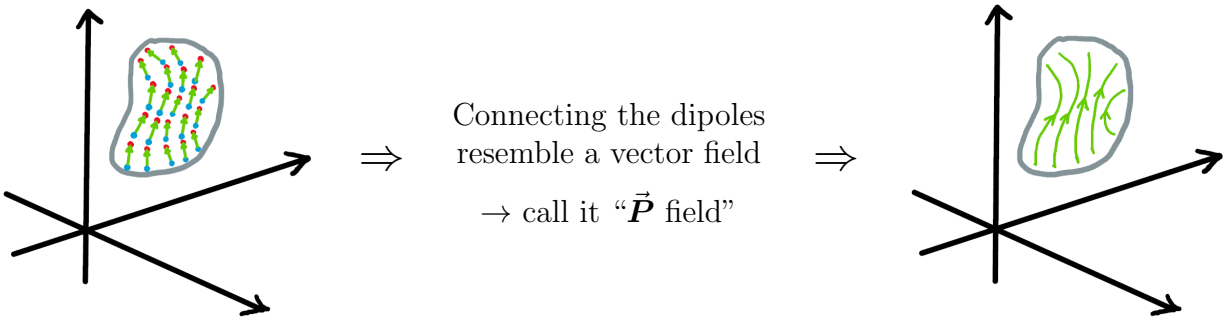
- **Polarization field** - $\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}})$
- **Bound charge distributions** - $\rho_b(\vec{\boldsymbol{r}})$ and $\sigma_b(\vec{\boldsymbol{r}})$

2.1 Polarization Field

When the dipoles in the material are so dense such that we can treat them as a continuous distribution, we can replace

- Sum of all dipoles $\xRightarrow{\text{become}}$ Volume integral over the whole object.
- Discrete dipoles source $\xRightarrow{\text{become}}$ A vector distribution called **polarization field** $\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}})$.

$$\boxed{V(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\iiint_{\substack{\text{Whole}\\\text{material}}}\frac{\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\cdot\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)\mathrm{d}^3\vec{\boldsymbol{r}}'}$$



The polarization field $\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}})$ can also be interpreted as **electric dipole density** because its usage is similar to charge density $\rho(\vec{\boldsymbol{r}})$. Comparing with the Coulomb's law for electric potential:

- When the source is made of charges:

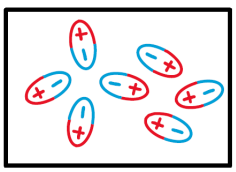
$$V(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\iiint \frac{\rho(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\,\mathrm{d}^3\vec{\boldsymbol{r}}' \quad\sim\quad \frac{1}{4\pi\epsilon_0}\sum \frac{(\text{charge density})}{(\text{distance})}$$

- When the source is made of dipoles:

$$V(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\epsilon_0}\iiint \frac{\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\cdot\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)\mathrm{d}^3\vec{\boldsymbol{r}}' \quad\sim\quad \frac{1}{4\pi\epsilon_0}\sum \frac{(\text{dipole density})}{(\text{distance})^2}\cdot\left(\begin{smallmatrix}\text{unit}\\ \text{vector}\end{smallmatrix}\right)$$

2.2 Bound Charge Distribution

Ultimately, electric dipoles are just pairs of charges. If the dipoles are aligned non-uniformly in a material, some regions may appear to have a higher density of positive heads than its surroundings, and so as negative tails.



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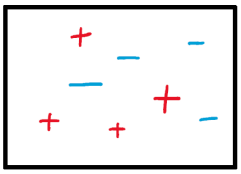


These regions with extra dipole heads/tails are described as **bound charges** - they are always "bounded" to regions where there are extra dipole heads/tails. With more vector calculus, we can derive their relation with $\vec{\boldsymbol{P}}$:

$$\begin{aligned}
V(\vec{\boldsymbol{r}}) &= \frac{1}{4\pi\epsilon_0}\iiint_{\substack{\text{Whole}\\ \text{material}}} \frac{\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\cdot\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)\mathrm{d}^3\vec{\boldsymbol{r}}' \\
&= (\cdots\ \textit{After more boring vector calculus}\ \cdots) \\
&= \frac{1}{4\pi\epsilon_0}\iiint_{\substack{\text{Whole}\\ \text{material}}} \frac{-\vec{\boldsymbol{\nabla}}\cdot\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\,\mathrm{d}^3\vec{\boldsymbol{r}}' + \frac{1}{4\pi\epsilon_0}\oiint_{\substack{\text{Surface}\\ \text{of material}}} \frac{\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\cdot\mathrm{d}^2\vec{\boldsymbol{r}}'
\end{aligned}$$

Notice that these two integrals are very similar to Coulomb's law of electric potential:

$$\begin{cases}
\dfrac{1}{4\pi\epsilon_0}\displaystyle\iiint_{\substack{\text{Whole}\\ \text{material}}} \frac{-\vec{\boldsymbol{\nabla}}\cdot\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\,\mathrm{d}^3\vec{\boldsymbol{r}}' \quad\sim\quad \dfrac{1}{4\pi\epsilon_0}\displaystyle\sum_{\substack{\text{Inside}\\ \text{material}}} \frac{(-\vec{\boldsymbol{\nabla}}\cdot\vec{\boldsymbol{P}})}{(\text{distance})} \\[2ex]
\dfrac{1}{4\pi\epsilon_0}\displaystyle\oiint_{\substack{\text{Surface}\\ \text{of material}}} \frac{\vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\cdot\mathrm{d}^2\vec{\boldsymbol{r}}' \quad\sim\quad \dfrac{1}{4\pi\epsilon_0}\displaystyle\sum_{\substack{\text{On material}\\ \text{surface}}} \frac{(\text{Flux of }\vec{\boldsymbol{P}})}{(\text{distance})}
\end{cases}$$

What if we interpret these two as charges?

Their compositions suggest that we can define two new charge densities expressions:

- **Volume bound charge density**

$$\boxed{\rho_b(\vec{\boldsymbol{r}}) \stackrel{\text{def}}{=} -\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{P}}(\vec{\boldsymbol{r}})}$$

- **Surface bound charge density**

$$\boxed{\sigma_b(\vec{\boldsymbol{r}}) \stackrel{\text{def}}{=} \vec{\boldsymbol{P}}(\vec{\boldsymbol{r}}) \cdot \hat{\boldsymbol{n}}}$$

  Here the $\hat{\boldsymbol{n}}$ is the (outward) unit normal vector on the material surface. $\vec{\boldsymbol{P}} \cdot \hat{\boldsymbol{n}}$ means we are taking the out-flux of $\vec{\boldsymbol{P}}$.

Then the potential formula can be interpreted as:

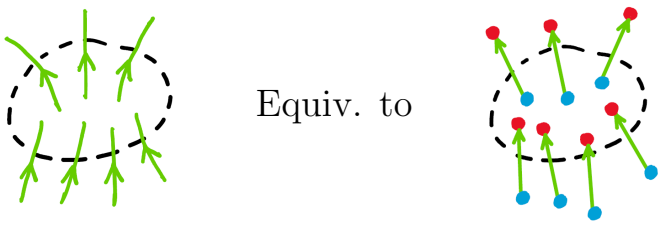
$$\begin{aligned} V(\vec{\boldsymbol{r}}) &= \frac{1}{4\pi\epsilon_0} \iiint\limits_{\substack{\text{Whole}\\ \text{material}}} \frac{\rho_b(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}'|} \, \mathrm{d}^3\vec{\boldsymbol{r}}' + \frac{1}{4\pi\epsilon_0} \oiint\limits_{\substack{\text{Surface}\\ \text{of material}}} \frac{\sigma_b(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}'|} \, \mathrm{d}^2\vec{\boldsymbol{r}}' \\ &= \begin{pmatrix} \text{Contribution} \\ \text{by charges inside} \\ \text{the material} \end{pmatrix} + \begin{pmatrix} \text{Contribution} \\ \text{by charges on} \\ \text{material's surface} \end{pmatrix} \end{aligned}$$

which is simply saying that the electric potential from a dielectric is contributed by two distributions of charges - one through out the whole material and one only appears on the surface.

2.2.1 Volume Bound Charge Density

Consider drawing a closed surface inside the $\vec{\boldsymbol{P}}$ field,

More $\vec{\boldsymbol{P}}$ field pointing into the surface	$\Leftrightarrow$	More positive charge contained inside the surface



Recall what we have learnt about Gauss's law:

- We can use sign of flux to identify positions with converging/diverging vector field.
- Divergence operator $\vec{\boldsymbol{\nabla}}\cdot$ is equivalent to finding the out-flux per volume of a vector field.

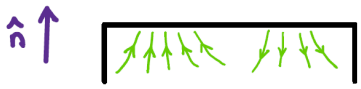
This is how we can interpret the definition:

$$\begin{pmatrix} \text{Bound charge} \\ \text{volume density} \end{pmatrix} \sim \rho_b = -\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{P}} \sim \begin{pmatrix} \text{In-flux of dipole} \\ \text{per volume} \end{pmatrix}$$

2.2.2 Surface Bound Charge Density

Consider where the $\vec{\boldsymbol{P}}$ field terminates at the material surface,

More $\vec{\boldsymbol{P}}$ field pointing the same direction as $\hat{\boldsymbol{n}}$	$\Leftrightarrow$	More positive charge on the material surface



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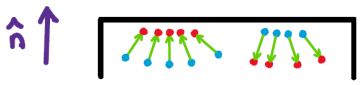


Since we choose the normal vector $\hat{\boldsymbol{n}}$ to be pointing outward of the material's surface,

- The more dipoles in similar direction as $\hat{\boldsymbol{n}}$, the more positive charges on the surface.
- We can use dot product to check if two vectors are in the similar direction.

Therefore surface bound charge density involve dot product with $\hat{\boldsymbol{n}}$.

$$\begin{pmatrix}\text{Bound charge}\\ \text{surface density}\end{pmatrix} \sim \sigma_b = \vec{\boldsymbol{P}} \cdot \hat{\boldsymbol{n}} \sim \begin{pmatrix}\text{Out-flux of dipole}\\ \text{per surface area}\end{pmatrix}$$

The boring derivation of the 2 bound charge terms:

1. First we need to show that the unit vector part is related to gradient:

Note: Differentiation is w.r.t. x', y', z'

$$\begin{aligned}
\vec{\boldsymbol{\nabla}}\left(\frac{1}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right) &= \left(\hat{\boldsymbol{x}}\frac{\partial}{\partial \underline{x'}} + \hat{\boldsymbol{y}}\frac{\partial}{\partial \underline{y'}} + \hat{\boldsymbol{z}}\frac{\partial}{\partial \underline{z'}}\right)\left(\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\right) \\
&= \frac{\hat{\boldsymbol{x}}(x-x') + \hat{\boldsymbol{y}}(y-y') + \hat{\boldsymbol{z}}(z-z')}{[(x-x')^2+(y-y')^2+(z-z')^2]^{\frac{3}{2}}} \\
&= \frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^3} \\
&= \frac{1}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\underbrace{\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)}
\end{aligned}$$

This part is just a unit vector in $\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'$ direction

This is basically $\frac{\mathrm{d}}{\mathrm{d}r'}\left(\frac{1}{r-r'}\right) = \frac{1}{(r-r')^2}$ but in vector calculus version. An additional unit vector is present because result of gradient must be a vector field.

2. Substitute into the potential formula to become:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\iiint \underbrace{\frac{\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|^2}\cdot\left(\frac{\vec{r}-\vec{r}\,'}{|\vec{r}-\vec{r}\,'|}\right)}_{\text{Result from step 1}} \mathrm{d}^3\vec{r}\,'$$

$$= \frac{1}{4\pi\epsilon_0}\iiint \underbrace{\vec{P}(\vec{r}\,')\cdot\vec{\nabla}\left(\frac{1}{|\vec{r}-\vec{r}\,'|}\right)}_{\vec{G}\cdot\vec{\nabla}f} \mathrm{d}^3\vec{r}\,'$$

$$= \frac{1}{4\pi\epsilon_0}\iiint \underbrace{\vec{\nabla}\cdot\left(\frac{\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\right)}_{\vec{\nabla}\cdot(f\vec{G})} - \underbrace{\frac{\vec{\nabla}\cdot\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}}_{-f(\vec{\nabla}\cdot\vec{G})} \mathrm{d}^3\vec{r}\,'$$

Here we used the **product rule of divergence** from line 2 to 3,:

$$\boxed{\vec{\nabla}\cdot(f\vec{G}) = f(\vec{\nabla}\cdot\vec{G}) + (\vec{\nabla}f)\cdot\vec{G}}$$

where f is a scalar function (like $\frac{1}{|\vec{r}-\vec{r}\,'|}$) and $\vec{G}$ is a vector function (like $\vec{P}$).

3. Finally use divergence theorem to convert the 1st term's volume integral into a surface integral over the volume's surface:

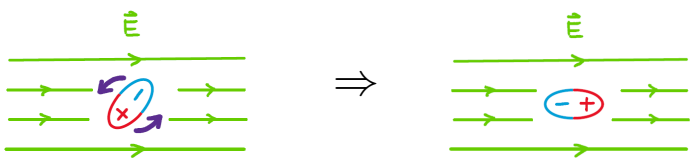
$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\underbrace{\iiint \vec{\nabla}\cdot\left(\frac{\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\right)}_{\text{Divergence theorem}} \mathrm{d}^3\vec{r}\,' - \frac{1}{4\pi\epsilon_0}\iiint \frac{\vec{\nabla}\cdot\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\mathrm{d}^3\vec{r}\,'$$

$$= \frac{1}{4\pi\epsilon_0}\underbrace{\oiint \frac{\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\cdot \mathrm{d}^2\vec{r}\,'} - \frac{1}{4\pi\epsilon_0}\iiint \frac{\vec{\nabla}\cdot\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\mathrm{d}^3\vec{r}\,'$$

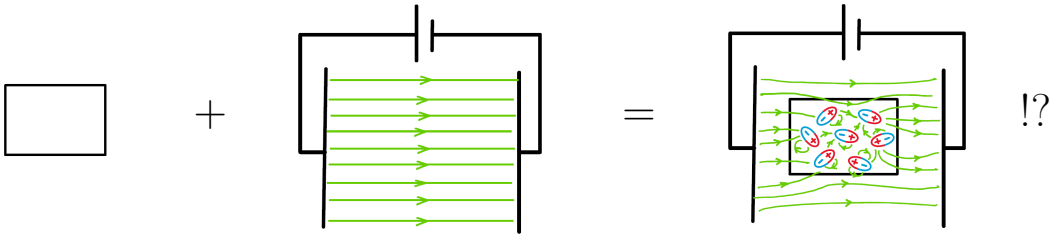
$$= \begin{pmatrix}\text{Contribution}\\ \text{from material's}\\ \text{surface}\end{pmatrix} + \begin{pmatrix}\text{Contribution}\\ \text{from inside}\\ \text{the material}\end{pmatrix}$$

3 Material under External E-field

If we place a single electric dipole under an external E-field, the dipole will rotate itself until it is in the same direction as the E-field.



However in a material structure, bondings/interactions exist between dipoles, which prevent the dipoles from freely rotating. So the final dipole alignment might not always align with the external E-field!



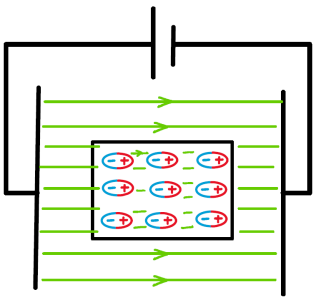
In reality, fully model a material's response to E-field can be very complicated. The general theory assumes the polarization field $\vec{\boldsymbol{P}}$ (i.e. new dipole alignment) as some function to the total E-field $\vec{E}_{\text{total}}$:

$$\vec{\boldsymbol{P}} \;=\; f(\vec{\boldsymbol{E}}_{\text{total}}) \;\sim\; \underbrace{\vec{\boldsymbol{a}}_i^{(1)} E_i + \vec{\boldsymbol{a}}_{ij}^{(2)} E_i E_j + \vec{\boldsymbol{a}}_{ijk}^{(3)} E_i E_j E_k + \cdots}_{\text{Like a Taylor expansion}}$$

This f function denotes a theoretical model that we have choosen to investigate the material. And the model parameters $\vec{\boldsymbol{a}}^{(1)}, \vec{\boldsymbol{a}}^{(2)}, \vec{\boldsymbol{a}}^{(3)}$... shall be determined from experiment.

3.1 Special Case: Linear Dielectric

Linear dielectric is the simplest model of E-field reponse - where the dipoles are completely free to rotate, such that new dipole alignment is directly proportional to the external E-field's magnitude and direction. This assumption is applicable to most daily life materials.



No random arrangement.
Every dipole is well-aligned.

The linear model only has 1 parameter to be determined from experiments - the proportionality constant between $\vec{\boldsymbol{P}}$ and $\vec{\boldsymbol{E}}_{\text{total}}$.

$$\boxed{\vec{\boldsymbol{P}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}}_{\text{total}}}$$

Here the χ_e is called **electric susceptibility**, a pure number (no unit) whose value depends on the type of material. ϵ_0 is multiplied to match the units.

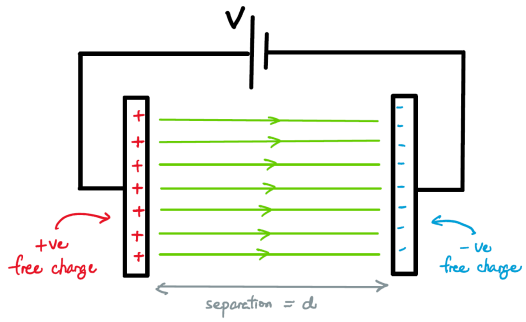
- $\chi_e = 0$ for vacuum (by defintion), because there are no dipoles in a vacuum.
- $\chi_e > 0$ for most materials, because dipole moments normally follow E-field's direction.

3.2 Describing External Field

The total E-field is a result of the external E-field plus the field induced by dipole alignment. After we have chosen the model about dipole alignment, now we look at the external E-field.

3.2.1 Free Charge Distribution

In order to create an external E-field, we need a "setup" to build an external source of charges. For example, apply voltage between two very large parallel plates:



The charges that build up in the setup are given the name **free charges**, to distinguish from the bound charges (dipole alignment) in the material. The are "free" because we can always control them by varying the setup, making them known quantities in calculation.

$$\text{The actual thing we can control} = \text{Voltage} \longrightarrow \frac{V}{d} = |\vec{\boldsymbol{E}}| = \frac{\sigma_{\text{free}}}{\epsilon_0} \longleftarrow \text{Free charge density depends on applied voltage}$$

For calculation, free charge densities are denoted like bound charges densities:

- **Volume free charge density** $= \rho_f(\vec{\boldsymbol{r}})$ - Charge distribution inside the setup.
- **Surface free charge density** $= \sigma_f(\vec{\boldsymbol{r}})$ - Charge on any surfaces of the setup.

3.2.2 Displacement Field $\vec{\boldsymbol{D}}$

Previously, we have relate bound charges to a vector field quantity - the polarization field $\vec{\boldsymbol{P}}$. Similarly, free charges can be related with another vector field. From Gauss's law,

$$\begin{aligned}
\epsilon_0 \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{E}}_{\text{total}} &= \rho_{\text{total}} = \begin{pmatrix}\text{All}\\ \text{charges}\end{pmatrix} \\
&= \rho_f + \rho_b = \begin{pmatrix}\text{Free charges}\\ \text{on setup}\end{pmatrix} + \begin{pmatrix}\text{Bound charges}\\ \text{on material}\end{pmatrix} \\
&= \rho_f + (-\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{P}}) \\
\vec{\boldsymbol{\nabla}} \cdot \left(\epsilon_0 \vec{\boldsymbol{E}}_{\text{total}} + \vec{\boldsymbol{P}}\right) &= \rho_f
\end{aligned}$$

Here we define the **displacement field** $\vec{\boldsymbol{D}}$:

$$\boxed{\vec{\boldsymbol{D}} \stackrel{\text{def}}{=} \epsilon_0 \vec{\boldsymbol{E}}_{\text{total}} + \vec{\boldsymbol{P}}} \longleftarrow \text{This relation connects all 3 field quantities}$$

Such that it is related to the free charge density by:

$$\boxed{\begin{aligned} \rho_f(\vec{\boldsymbol{r}}) &\overset{\text{def}}{=} \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{D}}(\vec{\boldsymbol{r}}) \\ \sigma_f(\vec{\boldsymbol{r}}) &\overset{\text{def}}{=} \vec{\boldsymbol{D}}(\vec{\boldsymbol{r}}) \cdot \hat{\boldsymbol{n}} \end{aligned}}$$

Here the $\hat{\boldsymbol{n}}$ is the (outward) unit normal vector on the equipment surface. Dot product with it to represent out-flux.

Notice the relations of $\vec{\boldsymbol{D}}$ with free charges are almost identical to $\vec{\boldsymbol{E}}$ with total charges in Gauss's law. In the special case of linear dielectric, we can solve for $\vec{\boldsymbol{D}}$ from the given free charges like the Gauss's law integral form or with Coulomb's law.

$$\begin{aligned} Q_f &= \oiint \vec{\boldsymbol{D}}(\vec{\boldsymbol{r}}) \cdot \mathrm{d}\vec{\boldsymbol{s}} \\ &= \oiint \underline{|\vec{\boldsymbol{D}}||\,\mathrm{d}\vec{\boldsymbol{s}}\,|\cos\theta} \quad \longleftarrow \text{Just dot product } \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta \\ &= \underline{|\vec{\boldsymbol{D}}|}\ \underline{\cos\theta} \oiint |\,\mathrm{d}\vec{\boldsymbol{s}}\,| \\ &= |\vec{\boldsymbol{D}}|\ \cos\theta\ (\text{Total surface area}) \end{aligned}$$

Same magnitude everywhere
Can move out of integral

Form same angle everywhere
Can move out of integral

$$\boxed{|\vec{\boldsymbol{D}}| = \frac{Q_f}{(\text{Total surface area})\cos\theta}}$$

But note that this is incorrect for other models of dielectrics.

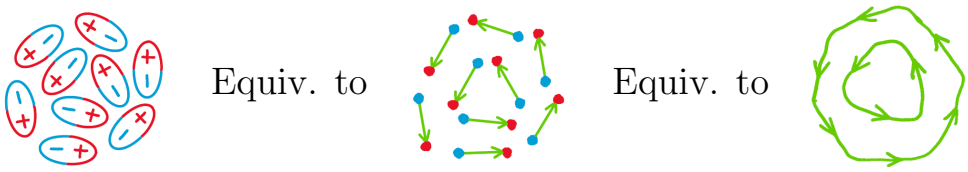
Side Note:

In electrostatics, if we know the total charge distribution, we can solve for the total E-field uniquely by one PDE (Gauss's Law). However we cannot do this to $\vec{\boldsymbol{D}}$ even though it relates to free charge by a similar PDE.

$$\boxed{\begin{array}{c} \text{Solve } \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{E}}_{\text{total}} = \dfrac{\rho_{\text{total}}}{\epsilon_0} \\ \Downarrow \\ \vec{\boldsymbol{E}}_{\text{total}} \text{ has a unique solution} \end{array}} \quad \textbf{BUT} \quad \boxed{\begin{array}{c} \text{Solve } \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{D}} = \rho_f \\ \Downarrow \\ \vec{\boldsymbol{D}} \text{ still has many possibilities} \end{array}}$$

This is because $\vec{\boldsymbol{E}}_{\text{total}}$ and $\vec{\boldsymbol{D}}$ are different in their "curl-ness":

- $\vec{\boldsymbol{E}}_{\text{total}}$ is guarenteed to be curl $= 0$ in electrostatics, i.e. E-field lines never form loops. So we can define a potential function V to solve the PDE uniquely.
- $\vec{\boldsymbol{P}}$ can have non-zero curl. For example, if the dipoles form some vortex-like pattern under external E-field.
- Then by $\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}}_{\text{total}} + \vec{\boldsymbol{P}}$, $\vec{\boldsymbol{D}}$ cannot be guarenteed to be curl $= 0$.



($\vec{\boldsymbol{E}}$ cannot form rotating pattern, but $\vec{\boldsymbol{P}}$ can. So does $\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}} + \vec{\boldsymbol{P}}$.)

Therefore we have to know the material's model before working with $\vec{\boldsymbol{D}}$ and $\vec{\boldsymbol{P}}$. For example, in the special case of linear dielectric,

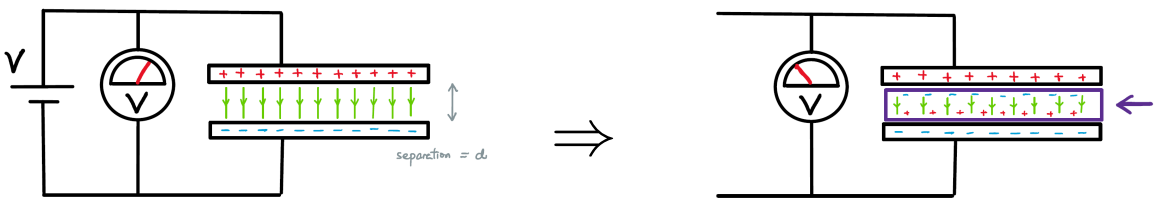
$$\boxed{\vec{\boldsymbol{P}} \text{ aligns with } \vec{\boldsymbol{E}}_{\text{total}}} \Rightarrow \boxed{\vec{\boldsymbol{P}} \text{ is "curl-less"}} \Rightarrow \boxed{\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}}_{\text{total}} + \vec{\boldsymbol{P}} \text{ is also "curl-less"}}$$

So we are safe to solve for $\vec{\boldsymbol{D}}$ like normal Gauss's law problems ONLY after we know it is the linear model.

3.3 Measurement in Practice

In experiments that involve dielectric,

- It is impossible to measure $\vec{\boldsymbol{P}}$ - We cannot put any sensors inside a material without breaking it. The best we can do is to choose a good theoretical model that fits the material's atomic arrangement. *(E.g. Given a linear dielectric ...)*
- $\vec{\boldsymbol{D}}$ is control-able - $\vec{\boldsymbol{D}}$ is related to free charge, and free charge can be prepared by initially charging up the setup to a desirable voltage.
- $\vec{\boldsymbol{E}}_{\text{total}}$ is measure-able - Since $\vec{\boldsymbol{E}} \propto (\vec{\boldsymbol{D}} - \vec{\boldsymbol{P}})$, we expect a difference in $\vec{\boldsymbol{E}}$ before and after placing the material sample in the setup. This change in $\vec{\boldsymbol{E}}$ can be measured by voltmeter.



Charge the setup to some voltage V.
Free surface charge $\sim \frac{\epsilon_0 V}{d}$

Remove power supply, insert material.
Change in voltmeter reading $\propto$ Change in $\vec{\boldsymbol{E}}$

Once we have chosen a model of the material, we can connect the two measurables by:

$$\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}}_{\text{total}} + f(\vec{\boldsymbol{E}}_{\text{total}})$$

Then the model parameters in $f(\cdots)$ can be determined by experiments. In the special case of linear dielectric, since there is only 1 constant parameter χ_e for each material,

$$\begin{aligned}\vec{\boldsymbol{D}} &= \epsilon_0 \vec{\boldsymbol{E}}_{\text{total}} + \epsilon_0 \chi_e \vec{\boldsymbol{E}}_{\text{total}} \\ &= \epsilon_0 (1 + \chi_e) \vec{\boldsymbol{E}}_{\text{total}} \\ &= (\text{A constant}) \cdot \vec{\boldsymbol{E}}_{\text{total}}\end{aligned}$$

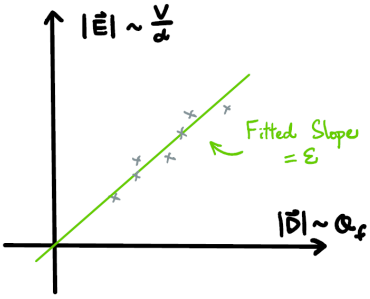
We usually use this new constant to represent the material's response to E-field, by defining

$$\boxed{\epsilon \overset{\text{def}}{=} \epsilon_0(1+\chi_e) \overset{\text{def}}{=} \epsilon_0\epsilon_r} \qquad \text{such that} \qquad \vec{\boldsymbol{D}} = \epsilon \vec{\boldsymbol{E}}_{\text{total}}$$

where

- ϵ = **Absolute permittivity**, or **dielectric constant**. Has the same unit as ϵ_0.
- ϵ_r = **Relative permittivity**. A pure number that normally > 1.

Material scientists prefer using ϵ instead of χ_e because of convenience - since $\vec{\boldsymbol{D}}$ and $\vec{\boldsymbol{E}}_{\text{total}}$ are the measureables (through free charge and voltage), the slope in a $|\vec{\boldsymbol{D}}|$ v.s. $|\vec{\boldsymbol{E}}_{\text{total}}|$ graph immediately gives the value of ϵ. *Too lazy to convert it back to* χ_e.

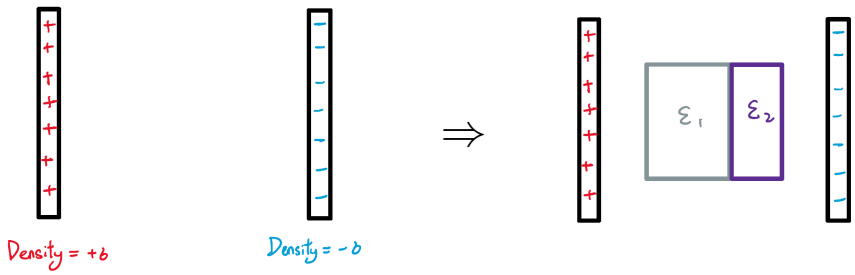


Material	Relative Permitivity (ϵ_r)
Vacuum	1
Oxygen	1.00049
Diamond	5.7
Salt (NaCl)	5.9
Iron rust (Fe_2O_3)	12
Ethanol	25.3
Water	80.1

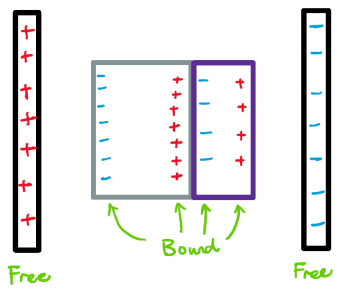
3.4 Calculation Example

Here are some examples of question problems you may see in the chapter of linear dielectric. In principle, one of the information about $\vec{\boldsymbol{D}}$ (free charge) or $\vec{\boldsymbol{E}}_{\text{total}}$ (voltage) will be given, then you are asked to find the other, and so as $\vec{\boldsymbol{P}}$ and bound charges.

Example 3.1. Consider two slabs of linear materials, each with <u>absolute</u> permittivity ϵ_1, ϵ_2, being placed between two large parallel plates. The plates are initially charged up to a surface density $\pm\sigma$ before putting in the materials. Assume the plates to be large enough such that the charged density and total E-field are uniform.



1. A good start is to label all the charges' locations, and if they are free/bounded.



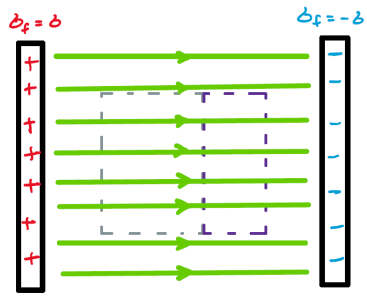
On setup = free

On material = bound

2. Always remember that $\vec{\boldsymbol{D}}$ is only related to the free charges. In this configuration, $\pm\sigma$ is the exact amount of free charge density on the plates since they are the manually placed in. We can treat it like the material does not exist.

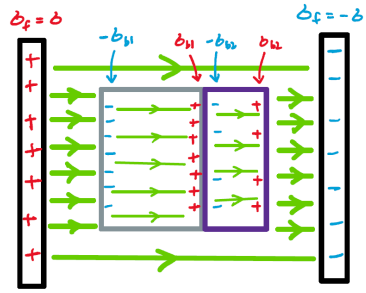
$$|\vec{\boldsymbol{D}}| = \sigma_f = \sigma$$

is the same everywhere between the plates.



3. According to the linear dielectric model, E-field is proportional to the $\vec{\boldsymbol{D}}$ by the dielectric constant at that location, i.e. $\vec{\boldsymbol{D}} = \epsilon\vec{\boldsymbol{E}}$. Because we have the same $\vec{\boldsymbol{D}}$ everywhere,

   - In vacuum, $|\vec{\boldsymbol{E}}| = \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon_0} = \dfrac{\sigma}{\epsilon_0}$
   - In material 1, $|\vec{\boldsymbol{E}}| = \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon_1} = \dfrac{\sigma}{\epsilon_1}$
   - In material 2, $|\vec{\boldsymbol{E}}| = \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon_2} = \dfrac{\sigma}{\epsilon_2}$

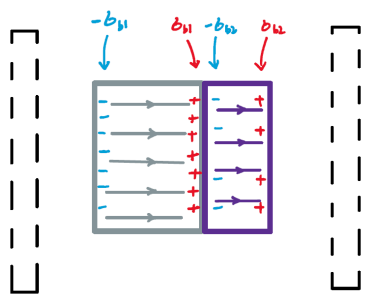


4. Then we can find $\vec{\boldsymbol{P}}$ by $\vec{\boldsymbol{P}} = \vec{\boldsymbol{D}} - \epsilon_0\vec{\boldsymbol{E}}$:

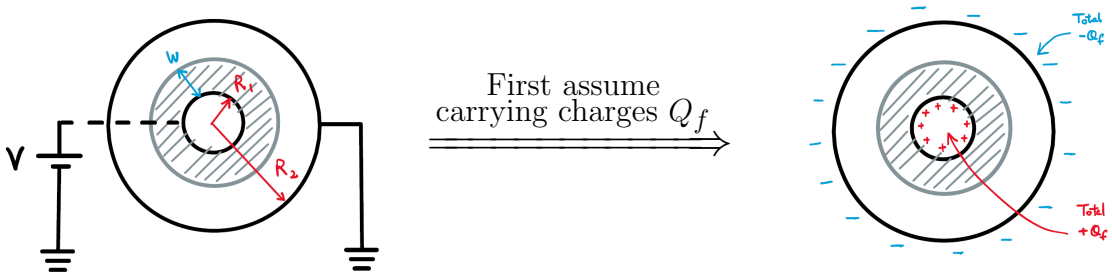
   - In vacuum, $|\vec{\boldsymbol{P}}| = |\vec{\boldsymbol{D}}| - |\vec{\boldsymbol{D}}| = 0$ (Obviously)
   - In material 1, $|\vec{\boldsymbol{P}}| = |\vec{\boldsymbol{D}}| - \epsilon_0 \cdot \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon_1} = \left(1 - \dfrac{\epsilon_0}{\epsilon_1}\right)\sigma$
   - In material 2, $|\vec{\boldsymbol{P}}| = |\vec{\boldsymbol{D}}| - \epsilon_0 \cdot \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon_2} = \left(1 - \dfrac{\epsilon_0}{\epsilon_2}\right)\sigma$

   According to $\vec{\boldsymbol{P}} \cdot \hat{\boldsymbol{n}} = \sigma_b$, the $|\vec{\boldsymbol{P}}|$ on each material are also the amount of surface bound charge densities on each of the materials' surfaces. i.e.

   - In material 1, $\sigma_{b1} = \left(1 - \dfrac{\epsilon_0}{\epsilon_1}\right)\sigma$
   - In material 2, $\sigma_{b2} = \left(1 - \dfrac{\epsilon_0}{\epsilon_2}\right)\sigma$



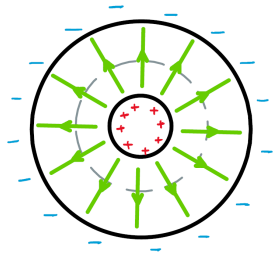
Example 3.2. Consider two spherical shells of radius R_1, R_2 connected by a voltage source V. The inner shell is covered by a layer of linear dielectric of constant ϵ and thickness w.



This time we are only given the applied voltage but not the free charge. However we can always tell that there must be some free charges on the shells. Let the total free charges to be $\pm Q_f$. **We can work backward after relating Q_f with V.**

1. By symmetry, $\vec{\boldsymbol{D}}$ must be uniform and radially outward. Again, we can treat it like the material does not exist when calculating $\vec{\boldsymbol{D}}$. Apply Gauss's law integral form for $\vec{\boldsymbol{D}}$:

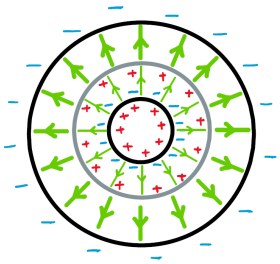
$$Q_f = \oiint \vec{\boldsymbol{D}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = |\vec{\boldsymbol{D}}| \cdot 4\pi r^2$$
$$\Rightarrow \quad |\vec{\boldsymbol{D}}| = \frac{Q_f}{4\pi r^2}$$



This is the $\vec{\boldsymbol{D}}$ everywhere between the shells, which is function to radial distance r.

2. According to the linear dielectric model, E-field is proportional to the $\vec{\boldsymbol{D}}$ by the dielectric constant at that location.

   - In the dielectric layer, $|\vec{\boldsymbol{E}}| = \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon} = \dfrac{Q_f}{4\pi\epsilon r^2}$
   - In the vacuum layer, $|\vec{\boldsymbol{E}}| = \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon_0} = \dfrac{Q_f}{4\pi\epsilon_0 r^2}$



3. Now we can find the free charge by relating the applied voltage V to the E-field by integrating along radial direction:

$$\begin{aligned} V &= \int \vec{\boldsymbol{E}} \cdot \mathrm{d}\vec{\boldsymbol{r}} \\ &= \int_{R_1}^{R_1+w} \frac{Q_f}{4\pi\epsilon r^2}\,\mathrm{d}r \;+\; \int_{R_1+w}^{R_2} \frac{Q_f}{4\pi\epsilon_0 r^2}\,\mathrm{d}r \\ &= \frac{Q_f}{4\pi\epsilon}\left(\frac{1}{R_1} - \frac{1}{R_1+w}\right) \;+\; \frac{Q_f}{4\pi\epsilon_0}\left(\frac{1}{R_1+w} - \frac{1}{R_2}\right) \end{aligned}$$

$$Q_f = \frac{4\pi V}{\frac{1}{\epsilon}\left(\frac{1}{R_1} - \frac{1}{R_1+w}\right) + \frac{1}{\epsilon_0}\left(\frac{1}{R_1+w} - \frac{1}{R_2}\right)}$$

4. Also $\vec{\boldsymbol{P}}$ by $\vec{\boldsymbol{P}} = \vec{\boldsymbol{D}} - \epsilon_0 \vec{\boldsymbol{E}}$:

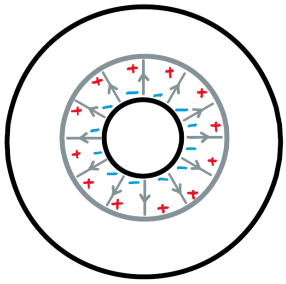
   - In the dielectric layer, $|\vec{\boldsymbol{P}}| = |\vec{\boldsymbol{D}}| - \epsilon_0 |\vec{\boldsymbol{E}}| = |\vec{\boldsymbol{D}}| - \epsilon_0 \cdot \dfrac{|\vec{\boldsymbol{D}}|}{\epsilon} = \left(1 - \dfrac{\epsilon_0}{\epsilon}\right) \dfrac{Q_f}{4\pi r^2}$
   - In the vacuum layer, $|\vec{\boldsymbol{P}}| = 0$ (Obviously)

5. This time $\vec{\boldsymbol{P}}$ is a function of r, so the surface bound charge density are different on the inner and outer surfaces.

   - On inner surface: $\sigma_b = |\vec{\boldsymbol{P}}|_{(r=R_1)} = \left(1 - \dfrac{\epsilon_0}{\epsilon}\right) \dfrac{Q_f}{4\pi R_1^2}$
   - On outer surface: $\sigma_b = |\vec{\boldsymbol{P}}|_{(r=R_1+w)} = \left(1 - \dfrac{\epsilon_0}{\epsilon}\right) \dfrac{Q_f}{4\pi (R_1+w)^2}$

   But the total amount of bound charge on each surface should still be the same (since the material is electrically neutral). This can be verified by:

$$
\begin{aligned}
(\text{Total charge}) &= \begin{pmatrix}\text{Surface}\\ \text{density}\end{pmatrix} \times \begin{pmatrix}\text{Surface}\\ \text{area}\end{pmatrix} \\
Q_b &= \sigma_b \cdot (4\pi r^2) \\
&= \left(1 - \frac{\epsilon_0}{\epsilon}\right) Q_f \\
&= (\text{Independent of } r)
\end{aligned}
$$



   We can also show that there is no volume bound charge in the material, although we must use its vector calculus definition.

$$
\begin{aligned}
\rho_b &= -\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{P}} \\
&= -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 |\vec{\boldsymbol{P}}| \right) \\
&= -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \left(1 - \frac{\epsilon_0}{\epsilon}\right) \frac{Q_f}{4\pi r^2} \right) \\
&= 0
\end{aligned}
$$

— The End —